# Brian **Lester**

*Toronto, CA*

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## Education

**University of Toronto** — *Toronto, Canada*
Ph.D. Student working with Colin Raffel. One year was at the University of North Carolina — *2022–Present*

**University of Pittsburgh** — *Pittsburgh, PA*
Double major, 3.51. **Computer Science** with Honors, 3.86. **Neuroscience**, 3.19. — *2012–2016*

**Shanghai American School** — *Shanghai, China*
High School, Team Lead in FIRST Robotics Competition. — *2008–2012*

## Selected Publications

**h-index: 12**

**The Power of Scale for Parameter-Efficient Prompt Tuning**
**Brian Lester**, Rami Al-Rfou, Noah Constant
***EMNLP 2021***, Online. **6380** citations

**Finetuned Language Models Are Zero-Shot Learners**
Jason Wei, Maarten Bosma, Vincent Y Zhao, Kelvin Guu, Adams Wei Yu, **Brian Lester**, Nan Du, Andrew M Dai, Quoc V Le
***ICLR 2022***, Online. **5614** citations

**SPoT: Better Frozen Model Adaptation through Soft Prompt Transfer**
Tu Vu, **Brian Lester**, Noah Constant, Rami Al-Rfou, Daniel Matthew Cer
***ACL 2022***, Dublin, Ireland and Online. **358** citations

**Scaling Up Models and Data with `t5x` and `seqio`**
Adam Roberts, Hyung Won Chung, Anselm Levskaya, Gaurav Mishra, James Bradbury, Daniel Andor, Sharan Narang, **Brian Lester**, Et Al. (8/42)
***JMLR 2023***. **232** citations

**Overcoming Catastrophic Forgetting in Zero-Shot Cross-Lingual Generation**
Tu Vu, Aditya Barua, **Brian Lester**, Daniel Matthew Cer, Mohit Iyyer, Noah Constant
***EMNLP 2022***, Abu Dhabi, UAE and Online. **82** citations

**An Effective Label Noise Model for DNN Text Classification**
Ishan Jindal, Daniel Pressel, **Brian Lester**, Matthew Nokleby
***NAACL 2019***, Minneapolis, Minnesota. **60** citations

**The Common Pile v0.1: An 8TB Dataset of Public Domain and Openly Licensed Text**
Nikhil Kandpal*, **Brian Lester***, Colin Raffel*, Et Al. (2/27)
***NeurIPS 2025***; *Datasets & Benchmarks*, San Diego, CA. **36** citations

**Git-Theta: A Git Extension for Collaborative Development of Machine Learning Models**
Nikhil Kandpal*, **Brian Lester***, Mohammed Muqeeth, Anisha Mascarenhas, Monty Evans, Vishal Baskaran, Tenghao Huang, Haokun Liu, Colin Raffel
***ICML 2023***, Honolulu Hawaii. **21** citations

**A Library for Rapid Modeling, Experimentation and Development of Deep Learning Algorithms targeting NLP**
Daniel Pressel, Sagnik Ray Choudhury, **Brian Lester**, Yanjie Zhao, Matt Barta
***ACL 2018***; *NLP-OSS Workshop*, Melbourne, Australia. **16** citations

**Realistic Evaluation of Model Merging for Compositional Generalization**
Derek Tam*, Yash Kant*, **Brian Lester***, Igor Gilitschenski, Colin Raffel
***TMLR 2026***, 2026. **19** citations

**Training LLMs over Neurally Compressed Text**
**Brian Lester**, Jaehoon Lee, Alex Alemi, Jeffrey Pennington, Adam Roberts, Jascha Sohl-Dickstein, Noah Constant
***TMLR 2024***. **21** citations

**Reducing Retraining by Recycling Parameter-Efficient Prompts**
**Brian Lester***, Joshua Yurtsever*, Siamak Shakeri, Noah Constant
*Preprint arXiv:2208.05577*, 2022. **14** citations

**Multiple Word Embeddings for Increased Diversity of Representation**
**Brian Lester**, Daniel Pressel, Amy Hemmeter, Sagnik Ray Choudhury, Srinivas Bangalore
*Preprint arXiv:2009.14394*, 2020. **11** citations

**Constrained Decoding for Computationally Efficient Named Entity Recognition Taggers**
**Brian Lester**, Daniel Pressel, Amy Hemmeter, Sagnik Ray Choudhury, Srinivas Bangalore
***EMNLP 2020***; *Findings*, Online. **12** citations

**iobes: Library for Span Level Processing**
**Brian Lester**
***ACL 2020***; *NLP-OSS Workshop*, Online. **8** citations

**TokSuite: Measuring the Impact of Tokenizer Choice on Language Model Behavior**

Gül Sena Altıntaş, Malikeh Ehghaghi, **Brian Lester**, Fengyuan Liu, Wanru Zhao, Marco Ciccone, Colin Raffel

***ICML 2026***, 2026. **11** citations

**Intent Features for Rich Natural Language Understanding**

**Brian Lester**, Sagnik Ray Choudhury, Rashmi Prasad, Srinivas Bangalore

***NAACL 2021***; *Industry Track*, Online.

# Work Experience

**Google DeepMind** *Mountain View, California*

Senior Research Engineer *2020–Present*

Deep Learning research with a focus on Natural Language Processing, large pre-trained models, and zero-shot transfer.

- Prompt Tuning: an efficient method of controlling large frozen pre-trained language models based on T5. Matches performance of full fine-tuning using only $0.003\%$ of the parameters. Open-sourced of our codebase and it has enabled $5$ published papers, $1$ product launch, and at least $3$ more in-flight papers.
- Flan: Multitask training for a 137 billion parameter transformer-based decoder-only language model to create a model that is more effective at zero-shot prompting and performs better using Prompt Tuning.
- SPoT: Using multitask prompts as strong initialization for Prompt Tuning resulting in increased performance. Also used prompt similarity to estimate task similarity and to predict transferability.
- Added partial network training, lazy loading, and pre-filling of the auto-regressive cache to t5x, the open-source reimplementation of T5 in Jax. This final change reduced inference latency from $30$ seconds to $2.4$.

**Interactions** *Ann Arbor, Michigan*

Machine Learning Engineer *2018–2020*

Built production grade deep learning solutions and lead research efforts to push the boundaries of performance.

- Designed novel neural network architectures for calibrated intent detection, slot filling, and named entity linking using ConvNets, bLSTM-CRFs, ranking models, and transformer-based seq2seq models.
- Designed label space, annotation guidelines, and data collection method for NLU component of dialogue systems.
- Created a cloud-native model training platform based on declarative pipelines and kubernetes. Built a deployment platform that powers NLU for multiple production dialogue systems.
- Built efficient, batched implementations of complex neural network architectures such as Beam Search. My CRF implementation reduced training time by a factor of $10$.

**Trove** *Ann Arbor, Michigan*

Lead Machine Learning Research Engineer *2017–2018*

Created a model training and serving platform that processed 200 million emails per day. Provided technical leadership to the ML team.

- Designed ConvNets for text classification to find sentences that contain questions. This powered a user-facing feature and was used to featurize the social graph created from email.
- Created neural ranking model was used to find coreferent mentions in the text and provide context to users.
- Used lexical features, as well as connectivity information in the email social graph, to identified bot accounts.

**Visteon Corporation** *Van Buren, Michigan*

Software Engineering Intern *2015*

Designed an adaptive system to minimize voice recognition errors based on ASR confidence scores. We patented this system and it is used in Mazda Cars.

# Presentations

**Prompt Tuning** *University of Michigan*, *University of North Carolina*

Deep Learning *2021–2022*

An overview of my work on Prompt Tuning, as well as our work—Flan and SPoT—directly built on Prompt Tuning. The talk includes a collection of insights about the behavior of soft prompts aggregated from others' followup work.

**NeurIPS Spotlight Talk on Mead-Baseline** *NeurIPS OSS Workshop*

Deep Learning *December 2018*

A spotlight talk, at the Open Source Software workshop at NeurIPS 2018, about our open-source toolkit, Mead-Baseline.

**Confidence and Calibration of Neural Network Models** *EMU ML Conference*

Deep Learning *March 2020* [1]

An overview of techniques used to adjust model calibration, evaluation of models that have the ability to "reject" decision with low confidence, and their uses in the NLU unit of a production dialogue system.

**Padding in Neural Networks for Natural Language Processing** *A2D-NLP*

Natural Language Processing *February 2020*

A survey of NLP building blocks with a focus on correctness and the need for padding in complex situations as well as places it is unexpected, like max-pooling following a 1D convolution.

**Optimization via NumPy and Cython** *Michigan Python Meetup*

Numerical Computation *January 2020*

I use a series of optimizations for computing pairwise Manhattan distance to introduce core NumPy concepts and Cython to reduce the runtime from multiple hours to just seconds.

**Input Representations of Deep Neural Networks** *PyData Ann Arbor*

Deep Learning *October 2017*

Using learned character-compositional input representations to create Deep Neural Networks with an open vocbaulary.

[1] Canceled due to COVID-19

# Skills

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|---|---|
| **Deep Learning** | Extensive experience building novel Neural Network architectures, generally for NLP. High-performance training with Data and Model Parallelism, including multihost distributed training on TPU. |
| **Infrastructure** | Build and deploy with Kubernetes, Docker, Flux, MongoDB, Apache Nifi, Github Actions, and GitLab CI/CD. Experience training large neural networks on Google Cloud (GCP) |
| **Toolkits** | Jax, Flax, PyTorch, Tensorflow, NumPy, Pandas, Faiss, SpaCy, NLTK, Tensorflow-Datasets, Seaborn, and Matplotlib. |
| **Languages** | Python, Cython, Java, C, Javascript, C++, Elisp, and LaTeX. |

# Service & Public Scholarship

## Reviewing

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|---|---|
| 2023–2025 | NeurIPS |
| 2022 | IEEE Transactions on Affective Computing |
| 2022 | ARR: ACL Rolling Review |
| 2022–2024 | NAACL |
| 2025 | ACL |
| 2025 | EMNLP |
| 2025 | ICLR |
| 2025 | ICML |
| 2020–2021 | Computer Speech and Language |
| 2019 | CoNLL |

## Public Scholarship

2020 [1] **a²-dlearn:** Helped organize logistics, recruit speakers, and acquire funding through sponsorships